

REV A Rework Guide — Boost Converter Fix

FLARM USB Updater | MT3608 boost produces 0V — L1 and D1 in wrong position | Fix: desolder L1/D1, resolder with correct topology

WRONG (REV A):

VUSB■IN SW■L1■VB00ST■D1■V12_OUT

CORRECT:

VUSB■L1■SW■D1■V12_OUT (C3, R1/R2)

Tools: Fine-tip iron, hot air (or 2nd iron), flux, 30AWG wire, tweezers, Kapton tape, multimeter, solder wick

Ref	Part	Package	Notes
L1	SMNR4020 22uH	4.0x4.0mm	Non-polarized
D1	B5819W SL	SOD-123	Cathode = bar marking

Step 1 — Desolder L1

- Apply flux around L1 (4x4mm shielded inductor)
- Heat with hot air ~350C or iron on both pads
- Lift off with tweezers, clean pads with wick
- Set L1 aside — will be reused

Step 2 — Desolder D1

- Apply flux around D1 (SOD-123)
- Heat with hot air or two irons simultaneously
- Lift off, clean pads, set aside for reuse

Step 3 — Bridge D1's empty pads

Solder blob or short wire across D1's two empty pads. Connects V12_OUT directly to VBOOST net.
[Verify: continuity RJ45 pin1/2 to C3+ pad](#)

Step 4 — Solder L1 with flying wires

- Cut two 30AWG wires ~15mm each
 - Solder one wire to each L1 pad (either way)
 - Wire A to **VUSB**: C1 positive pad
 - Wire B to **MT3608 pin 1 (SW)**: corner pin
- [Verify: L1-A to USB VBUS, L1-B to pin 1 only](#)

Step 5 — Solder D1 with flying wires

- Cut two 30AWG wires ~15mm each
 - Cathode = bar marking on package
 - Anode** wire to **MT3608 pin 1 (SW)**
 - Cathode** wire to **C3 positive pad**
- [Verify \(diode mode\): SW to C3+ reads 0.2-0.4V](#)

Step 6 — Secure components

- Tack L1 and D1 down with Kapton tape
- Check no wire-to-wire shorts
- Route wires away from other components

Pre-Power Verification Checklist

- No short between VUSB and GND
 - No short between V12_OUT and GND
 - Continuity: L1 wire A to VUSB rail
 - Continuity: L1 wire B to MT3608 pin 1
 - Continuity: D1 cathode to C3+ / V12_OUT
- Continuity: D1 anode to MT3608 pin 1
 - Diode check: SW to V12_OUT reads 0.2-0.4V fwd
 - D1 empty pads bridged: V12_OUT to C3+ continuity
 - No solder bridges on MT3608 (check all 6 pins)

Power-On Test

- Plug USB-C into a USB port (not charger)
- LED1 (green) should light up
- Measure V12_OUT at RJ45 pins 1&2 vs GND (4/7/8)
- Expected: ~12.4V**
- If OK, connect to FLARM and test update

Troubleshooting

Symptom	Check
0V on V12_OUT	D1 pads bridged? L1 to VUSB?
~5V on V12_OUT	D1 reversed (anode/cathode)
~5V and hot	Solder bridge on MT3608
Unstable voltage	Cold joint on L1 or D1 wires
USB port off	VUSB-GND short, check joints